

-- 1. CREATE: Create database and table

```
CREATE DATABASE company;
```

```
USE company;
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY AUTO_INCREMENT,  
    emp_name VARCHAR(50) NOT NULL,  
    department VARCHAR(30),  
    salary DECIMAL(10,2),  
    hire_date DATE  
);
```

-- 2. ALTER: Modify the table structure

```
ALTER TABLE employees ADD COLUMN email VARCHAR(100);
```

```
ALTER TABLE employees MODIFY COLUMN salary DECIMAL(12,2);
```

```
ALTER TABLE employees CHANGE COLUMN emp_name full_name VARCHAR(70);
```

```
ALTER TABLE employees ADD CONSTRAINT chk_salary CHECK (salary > 0);
```

-- 3. RENAME: Rename the table

```
RENAME TABLE employees TO staff;
```

-- 4. TRUNCATE: Remove all data but keep structure

TRUNCATE TABLE staff;

-- 5. DROP: Delete the table and database

DROP TABLE IF EXISTS staff;

DROP DATABASE company;

Outputs:

Creating a Table

```
mysql> CREATE TABLE employees (  
->     emp_id INT PRIMARY KEY AUTO_INCREMENT,  
->     emp_name VARCHAR(50) NOT NULL,  
->     department VARCHAR(30),  
->     salary DECIMAL(10,2),  
->     hire_date DATE  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql> desc employees;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| emp_id     | int           | NO   | PRI | NULL    | auto_increment |  
| emp_name   | varchar(50)   | NO   |     | NULL    |                |  
| department | varchar(30)   | YES  |     | NULL    |                |  
| salary     | decimal(10,2) | YES  |     | NULL    |                |  
| hire_date  | date          | YES  |     | NULL    |                |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.03 sec)
```

Altering a table

```
mysql> -- Add a constraint
mysql> ALTER TABLE employees ADD CONSTRAINT chk_salary CHECK (salary > 0);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc employees;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| emp_id     | int           | NO   | PRI | NULL    | auto_increment |
| full_name  | varchar(70)   | YES  |     | NULL    |                |
| department | varchar(30)   | YES  |     | NULL    |                |
| salary     | decimal(12,2) | YES  |     | NULL    |                |
| hire_date  | date          | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

Renaming a Table

```
mysql> RENAME TABLE employees TO staff;
Query OK, 0 rows affected (0.04 sec)

mysql> desc staff;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| emp_id     | int           | NO   | PRI | NULL    | auto_increment |
| full_name  | varchar(70)   | YES  |     | NULL    |                |
| department | varchar(30)   | YES  |     | NULL    |                |
| salary     | decimal(12,2) | YES  |     | NULL    |                |
| hire_date  | date          | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Drop

```
mysql> DROP TABLE IF EXISTS staff;
Query OK, 0 rows affected (0.04 sec)

mysql> show table;
ERROR 1064 (42000): You have an error in your
syntax to use near '' at line 1
mysql> drop database company;
Query OK, 0 rows affected (0.04 sec)
```